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Large Language Models and Machine Learning Framework for Predicting Dental Ceramics Performance

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ABSTRACT

Introduction and aims: Clinical fractures remain the primary cause of failure in dental all-ceramic restorations, highlighting the need to improve the mechanical performance and durability of ceramic material. This study aimed to develop a large language model (LLM)-based framework to automatically construct a structured database of dental ceramics and integrate it with machine learning (ML) to predict material properties and accelerate material design.

Methods: LLMs (Llama, Qwen, and DeepSeek) were employed to perform literature mining tasks, including text classification, information extraction from abstracts, and tabular data extraction. These processes were integrated into an automated pipeline to systematically extract and structure compositional and performance data from dental research articles. Ten ML algorithms were then trained using the curated database to establish predictive models of ceramic performance.

Results: In the classification task, a few-shot learning model with simple label prompts achieved an F1 score of 0.89. Fine-tuned LLMs achieved F1 scores exceeding 0.89 across various entity categories.ML models were developed to predict the classification of flexural strength, with the Extra Trees model performing best (F1 = 0.928), and external validation yielding F1 = 0.88. SHAP analysis identified ZrO_2 and SiO_2 as key contributor, and exhaustive search identified optimal compositional ranges.

Conclusions: This study demonstrates an AI-based pipeline combining LLM-driven data extraction and ML modelling, offering a scalable and accurate approach for accelerating the discovery and optimization of dental ceramics and other dental materials.

Clinical Relevance: The findings underscore the potential of advanced LLMs and ML models in restorative dentistry and materials research.

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Introduction

Dental all-ceramic restorative materials have excellent mechanical properties and biocompatibility, making them one of the most popular choices for clinical restorations.[1] These ceramics are composed of various chemical compositions, such as SiO_2, Li_2O, K_2O, and ZrO_2, and their properties, including flexural strength and wear resistance, have a direct effect on their performance and long-term durability in the oral cavity. Despite these advantages, clinical fractures are still the main cause of failure of all-ceramic restorations.[2] Given the relationship between composition and properties is often complex and nonlinear, predicting the behaviour of ceramic materials under clinical conditions is a challenging problem that needs to be solved. In recent years, machine learning (ML) techniques have shown breakthrough potential for material property prediction.[3-5] Recent research in dental materials has increasingly adopted data-driven approaches, such as machine-learning models designed to predict the performance of resin-based materials and dental adhesives,[6-8] as well as the clinical aesthetic outcomes of restorative materials.[9,10] By learning potential correlations between material composition and performance metrics, ML algorithms can substantially reduce the need for costly and time-consuming trial-and-error experiments.[11,12] However, the reliability of these models depends heavily on the comprehensiveness and accuracy of high-quality training datasets, and there is currently a lack of systematic composition-performance databases in the field of dental ceramics.

There are two main sources of scientific data: (1) results from researcher-initiated experiments or computational analyses, and (2) peer-reviewed data published in academic literature.[13] The abundance of experimental data available in the scientific literature could be used to develop accurate predictive models of material properties, which is essential for advancing ML applications in materials science. However, despite the richness of literature-based data, manual extraction that relies on domain expertise to process unstructured text from tens of thousands of publications is both time-consuming and labour-intensive. As the number of scientific publications continues to grow, it becomes increasingly difficult to maintain and update such databases manually.[14] Consequently, there is a growing need for efficient and accurate automated methods for literature data extraction.

Advances in natural language processing (NLP) technologies have created new opportunities for the automatic construction of large-scale material datasets. Low accuracy is a disadvantage of traditional rule-based NLP methods when it comes to dealing with complex semantic relationships.[15-19] The emergence of pretrained language models partially overcomes these limitations. Several domain-adapted models based on the BERT architecture, such as SciBERT,[20] MatSciBERT,[21] MaterialsBERT,[22] and PubMedBERT,[23] have shown promising performance in extracting material-related information. However, these models remain dependent on extensive annotated corpora for optimal performance. With the development of generative large language models (LLMs), conversational LLMs such as GPT-4 have demonstrated the ability to efficiently extract relevant data from large amounts of scientific literature through prompt engineering.[24] Furthermore, targeted fine-tuning enables LLMs to more effectively focus on materials knowledge and requirements, which can significantly improve the accuracy and robustness of material information extraction.[25] However, closed-source models raise concerns regarding data security, reproducibility, and computational cost.[25] These issues can be mitigated by employing open-source models that support local deployment, such as Llama, Qwen, and DeepSeek, which provide complementary strengths – Llama as a widely validated baseline,[26] Qwen for its strong multilingual adaptability,[27] and DeepSeek for its enhanced reasoning-oriented training.[28] Utilizing multiple open-source LLMs enables a more balanced and comprehensive evaluation of model performance. Within the field of dental research, the application of LLMs technology has primarily focused on the dental knowledge question-answering and disease diagnosis.[29-32] Nevertheless, the automated extraction of high-quality databases from the literature remains lacking in the field of dental materials.[33]

Thus, the objective of this study is to develop an automated data mining framework for dental ceramic materials by leveraging prompt engineering and fine-tuning strategies for LLMs. This framework is designed to automatically and accurately extract chemical composition and property data from a large body of literature, generating a structured and comprehensive database. Utilizing this database, we develop ML models to predict the properties of dental ceramic materials and identify important factors that affect their performance. The research hypothesis was that LLMs can accurately mine literature data and that subsequent ML analysis can reveal correlations between material composition and performance, thereby enabling reliable predictions of material properties. The integration of ML and data mining tools is anticipated to significantly accelerate the discovery and optimization of novel dental materials, offering a more efficient alternative to traditional trial-and-error approaches.

Methods

Literature search and data preprocessing

Relevant journal articles on dental ceramic materials were retrieved from the Web of Science and PubMed databases using the search process detailed in the Appendix. Literature was automatically downloaded in plain text or Excel format through the use of web robot (Figure 1A). A summary of the retrieved literature and its statistical distribution is presented in Appendix Figure 1. The programmatic retrieval of articles often contains a large amount of data such as DOI, article ID, title, journal name, etc., and we need to implement a corpus cleansing step to remove irrelevant metadata. Only essential elements, including the title, abstract, and DOI, were retained, and the cleaned data were converted into JSON format for downstream processing.

Literature classification

Abstracts retrieved from the literature were initially classified using LLMs. This classification step was prioritized as it enables efficient exclusion of irrelevant studies with minimal or no modification to the model architecture. Compared to performing full-scale information extraction on all retrieved

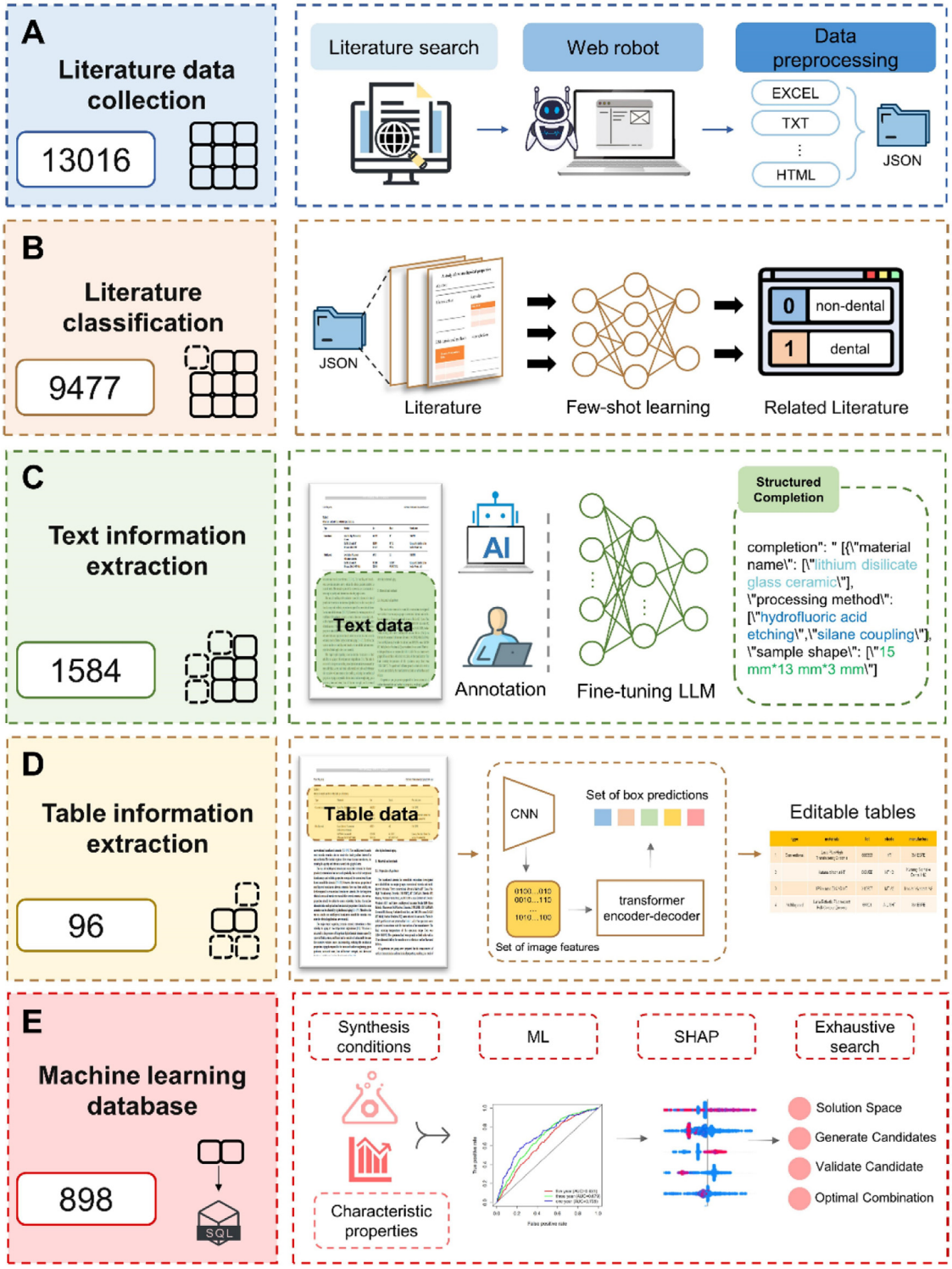


Figure 1 – Schematic overview of the automated literature mining pipeline. (A) Scientific articles are retrieved from online databases using a keyword-based search process. A web robot automatically downloads the documents, which are then pre-processed and converted from their original formats into structured JSON text. (B) Literature abstracts are categorized using LLMs under zero-shot learning, few-shot learning, and fine-tuning strategies, guided by both simple and complex label prompts. (C) Human-in-the-loop annotation is employed to generate labelled data, which is used to fine-tune LLMs for generating structured completions consistent with the annotation format. (D) An improved version of the table-transformer tool is used to extract data from scientific tables. (E) Predictive models are developed using machine learning approaches, accompanied by SHAP interpretability analysis and exhaustive compositional searches for material optimization.

articles, this approach significantly improves processing efficiency. Therefore, classification was adopted as a foundational step to support the subsequent extraction of composition–performance information (Figure 1B).[34] A total of 500 abstracts were manually annotated as either 'dental' or 'nondental'. To evaluate the text classification capabilities of LLMs, both zero-shot learning and few-shot learning approaches (using 10 labelled examples) were tested on models including Llama-3-70B-Instruct, Qwen2.5-72B-Instruct, and Deepseek-V3. The information regarding prompt engineering is included in the Appendix. Furthermore, the LLMs (Llama-2-70b-hf, Qwen2.5-72B-Instruct, and Deepseek-R1-Distill-Llama-70B) were fine-tuned using 400 labelled samples to assess their performance on the dental literature classification task.

Text information extraction

We fine-tuned the LLMs (Llama-2-70b-hf, Qwen2.5-72B-Instruct, and Deepseek-R1-Distill-Llama-70B) to perform the literature information extraction task using 523 manually annotated text extraction (prompt-completion) pairs (Figure 1C). A human–machine collaborative annotation strategy was employed for data labelling. Initially, domain experts refined preannotated outputs generated via the LLMs API. These annotations were then manually reviewed for accuracy, structured into a JSON format with defined logical relationships, and subsequently split into training and validation sets at a 9:1 ratio. A predefined standardization scheme was developed to format model inputs and outputs (Appendix Table 1), ensuring consistency with the structure of the training examples. Additional details regarding the annotation examples and the fine-tuning procedure are provided in the Appendix. In order to evaluate the information extraction performance of the LLMs, 10% of the data from each fold in the five-fold test set was sampled for manual evaluation. The evaluation criteria are outlined in the Appendix. In addition, to examine the generalizability of the fine-tuning models, we evaluated their performance in extracting information from dental adhesive literature using both zero-shot learning and fine-tuning approaches. Once fine-tuned on sufficient data adhering to the schema, a model will be capable of performing the same information extraction task on new text data with high accuracy. This enables the application of information extraction to data previously filtered through the classification step.

Table information extraction

In this study, we developed a deep learning–based pipeline for automated extraction of table information from the scientific literature (Figure 1D). Built upon an enhanced open-source table-transformer tool, the process comprises a three-stage framework: table detection and localization, table structure recognition, and functional analysis. Following optical character recognition (OCR), complete table content (table title and body) is converted into an editable, accurately recognized format. Relevant tables containing chemical composition and performance data are then identified through a screening step. The screening criteria are included in the Appendix.

Dataset construction and model training

A ML-ready database of dental ceramic materials was constructed following the information extraction process outlined above (Figure 1E). Over 10,000 articles were initially collected and screened, resulting in the inclusion of 96 publications from which 219 records of component-performance data were extracted. After excluding entries with incomplete performance values or ambiguous compositions, and averaging multiple test results for the same material, a final dataset comprising 60 distinct commercial dental ceramic brands was established (Appendix Table 2). Flexural strength values were categorized into five classes based on the ISO 6872:2015 standard. Given that the chemical composition data were predominantly reported as ranges, a data augmentation strategy was employed to enhance model training. This approach involved recursive traversal and Cartesian product generation, followed by random sampling and label assignment, to generate synthetic data that meets physical constraints. Using this method, a total of 898 training instances were produced for the development of the ML model.[35,36]

Ten ML algorithms were implemented using the scikit-learn package in Python (Appendix Table 3). Model training was conducted using 80% of the dataset, with the remaining 20% reserved for testing, to predict the flexural strength classification of ceramic materials. In the current study, the function 'Randomized Search CV' together with stratified 5-fold CV was used to search for the optimal hyperparameter combination for each model. Model performance was evaluated using multiple metrics, including precision, recall, and weighted F1 scores, supplemented by confusion matrices and precision-recall (PR) curves. PR curves, which plot precision against recall, are particularly advantageous over ROC curves for assessing models trained on imbalanced datasets.[37] To enhance interpretability, SHapley Additive exPlanation (SHAP) were used to analyse global and local feature contributions. To explore the material design space in depth, an exhaustive search strategy was developed. This involved multistage discrete sampling, combining uniform, extreme, and active sampling guided by feature importance to generate 480,565 hypothetical compositions.[38,39] The optimized ML model is then used to predict the flexural strength values of these virtual materials. The combination of descriptors with the highest and lowest predicted flexural strengths are compared.

Materials testing

To evaluate the generalizability of the model, six dental ceramic materials not included in the original database were selected as external test set. According to ISO 6872:2015, a total of 60 test specimens were fabricated based on the manufacturer's processing guidelines. Flexural strength was subsequently measured using a three-point bending test to assess the predictive performance of the ML model.

Results

We developed and implemented a framework for efficiently constructing a database from scientific literature (Figure 1).

The framework comprises multiple stages, including literature retrieval and preprocessing, LLM-based text classification and information extraction, table information extraction, and ML modelling. In the literature collection stage, 13,016 articles were retrieved from databases, and 12,866 documents were retained after deduplication and removal of incomplete records, providing a substantial corpus for downstream information extraction. Following preprocessing, the textual data were converted into structured JSON format to facilitate recognition and processing by LLMs.

We first evaluated the classification performance of the models using simple binary labels ('simple labels' of Figure 2A). Among the three open-source LLMs tested, Qwen achieved the highest F1 score (0.88), followed by Deepseek and Llama (Figure 2B). Subsequent evaluation under few-shot learning conditions showed performance improvements across all models, with Deepseek achieving the greatest gain – its F1 score increased from 0.87 to 0.89, representing the best result among the three models (Figure 2C). The fine-tuning models demonstrated a slight decline in performance, with the exception of

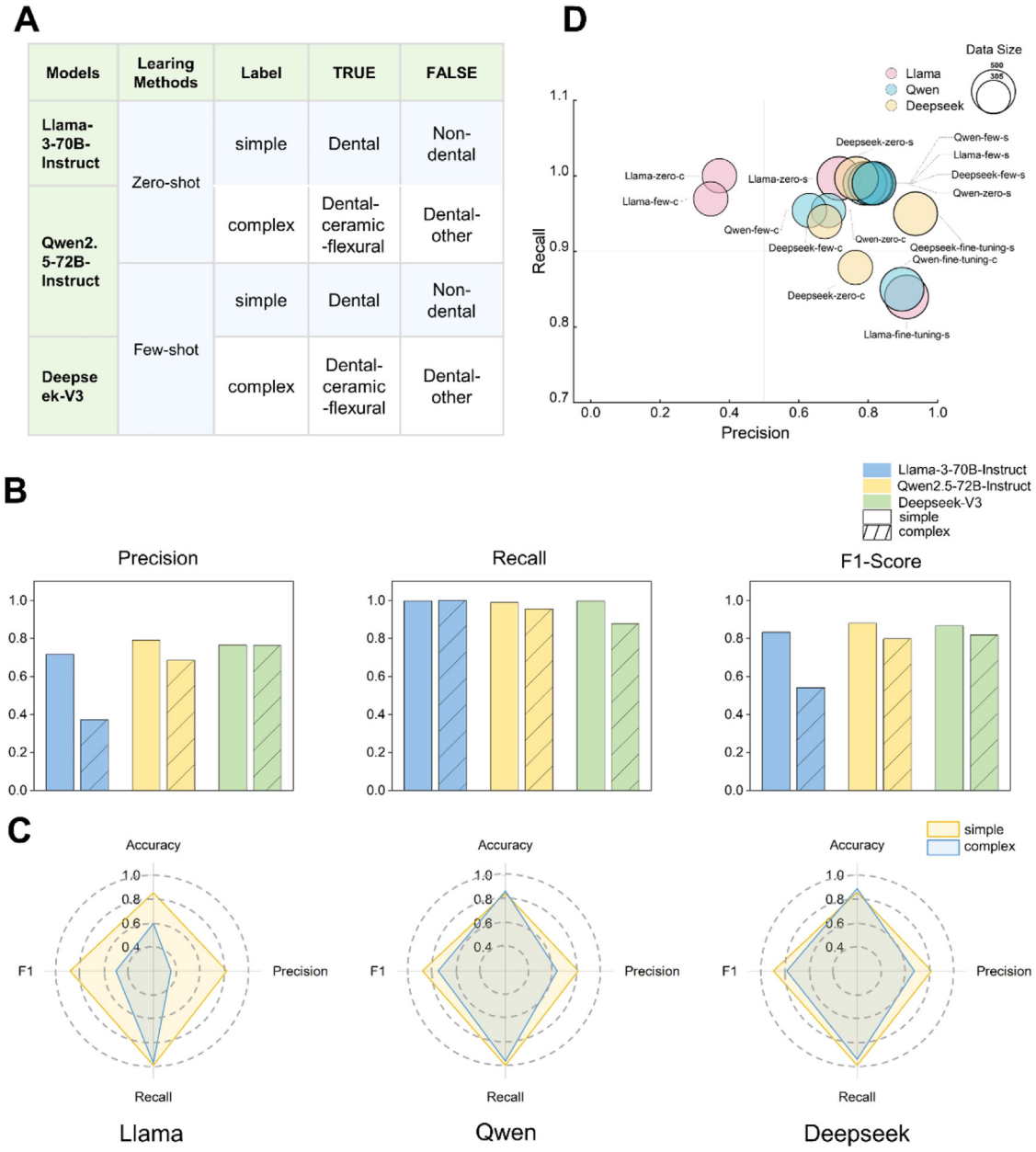


Figure 2 – Experimental results of the literature classification models. (A) Overview of classification models, label hierarchies, and definitions of label values. (B) Results of zero-shot learning with three open-source LLMs embeddings. The precision, recall, and F1 scores are reported. (C) Results of few-shot learning with three open-source LLMs embeddings. The precision, accuracy, recall, and F1 scores are reported. (D) Comparative performance of zero-shot learning, few-shot learning, and fine-tuning models. The x-axis and y-axis represent precision and recall, respectively. The node colour and size correspond to model types and dataset sizes.

the Deepseek model (Figure 2D). When tested in a classification task involving complex labels ('complex labels' of Figure 2A), and all models exhibited reduced performance under both zero-shot and few-shot learning settings (Appendix Tables 4 and 5). To balance model complexity and classification accuracy, the LLMs was employed in a few-shot learning setting with a simple label prompt strategy. This approach enabled the initial screening of 9477 articles relevant to the research scope.

In the task of extracting text information related to dental ceramic materials, results based on manual evaluation criteria showed that LLMs achieved F1 scores exceeding 0.89 across various entity categories (Figure 3A). To assess model generalizability, we evaluated the performance of fine-tuned ceramic models on dental adhesive literature entity recognition under a zero-shot learning setting. The Deepseek-R1-Distill-Llama-70B model exhibited consistently stable F1 scores

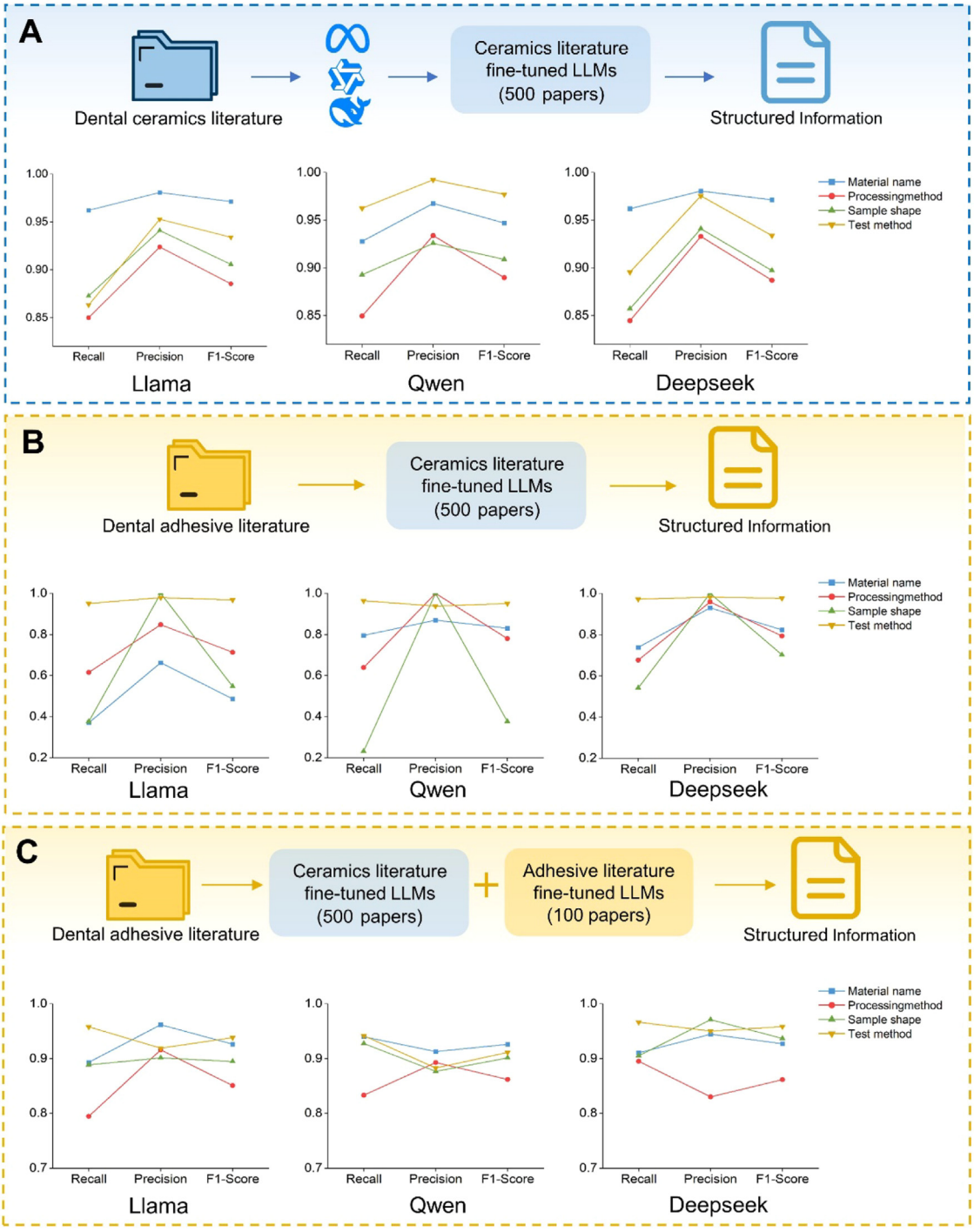


Figure 3 – Experimental results of text information extraction models. (A) Dental ceramic materials entity recognition performance comparison among three fine-tuning LLMs. (B) Dental adhesive materials entity recognition performance comparison among dental ceramic fine-tuning models. (C) Entity recognition performance for dental adhesive materials after fine-tuning the dental ceramic model with domain-specific data.

above 0.7 across all categories (Figure 3B), whereas the Llama-2-70b-hf and Qwen2.5-72B-Instruct models displayed more variability in performance across categories. Furthermore, we fine-tuned the ceramic model using only 100 manually annotated examples from the dental adhesive literature. The results showed that all LLMs achieved F1 scores exceeding 0.85 across various entity categories (Figure 3C). After classification stage, information extraction was conducted using a trained LLMs, resulting in the identification of 1584 articles based on predefined keyword criteria.

Tables, with their rich data and structured format, are optimal sources for information extraction. Through table parsing and screening, we identified 96 documents containing both component attributes and performance data that met the screening criteria. The statistical outcomes are shown in Appendix Figure 2.

Figure 4A demonstrates the three best-performing ML models for flexural strength classification, with the Extra Tree (ET) model achieving the highest predictive performance (Precision = 0.929, Recall = 0.923, F1-Score = 0.928). Detailed performance metrics for the remaining models were provided in the Appendix Table 6. Figure 4B illustrates the confusion matrix and PR curves used to evaluate classification performance. Figure 4C (i) shows the correlation between the variables in the heatmap. The global feature importance was analysed using SHAP, which revealed that the top 5 chemical components affecting the flexural strength were SiO_2 (feature importance score [SHAP_importance] = 0.11), ZrO_2 (0.09), B_2O_3 (0.05), Li_2O (0.03), Y_2O_3 (0.03) (Fig. 4C(ii)). Furthermore, SHAP interpretation of the fifth flexural strength category revealed opposing effects of SiO_2 and ZrO_2 on predicted values, and the same results were obtained in the analyses of the other classes as well (Figure 4C(iii)).

The ET model was applied to predict the flexural strength classifications of the virtual constituent compositions generated through exhaustive search. Over 100,000 samples from the virtual design space were visualized using the t-distributed Stochastic Neighbor Embedding (t-SNE) for dimensionality reduction (Figure 4D). Among them, compositions predicted with a probability of more than 80% for Class I and Class V flexural strength were identified. Further analysis revealed that high-ZrO_2 compositions were predominantly associated with Class V predictions, whereas those enriched in SiO_2 were more likely to be classified as Class I.

Figure 5A presents the stress–strain curves obtained from standardized three-point flexural strength testing of six ceramic materials comprising the external test set, along with corresponding data distribution plots. When applying the best-performing model to predict the flexural strength classifications of this external dataset, the ET model achieved an F1 score of 0.88. These results demonstrated the model's robust generalization ability and its reliability in predicting the flexural strength of dental ceramic materials (Figure 5B).

Discussion

The results of this study supported the research hypothesis that LLMs can accurately mine literature data and, when combined with ML, accurately predict material performance characteristics. The proposed framework demonstrated robust information extraction capabilities for dental ceramic literature and was successfully applied to extract and filter data from more than 10,000 documents. ML models were subsequently developed to predict the properties of dental ceramics. When validated against experimental data, the models achieved an F1 score of 0.88, indicating strong predictive performance. These findings underscore the potential of the LLM-integrated framework to advance automated information extraction and data mining in materials science.

Annotation for scientific literature information extraction is often a time-consuming and error-prone task. Incorporating a human-in-the-loop annotation strategy can significantly reduce the time required to collect the complete training set.[40] By checking and correcting annotations pregenerated by LLMs, the resulting annotations are of higher quality and faster compared to manual annotation from scratch. However, outputs generated by nonfine-tuned models still required manual correction by domain experts, indicating that baseline LLMs are not yet capable of reliably performing information extraction tasks in specialized domains. Therefore, appropriate fine-tuning strategies are essential to enhance the performance of LLMs in domain-specific applications.

In the classification task using LLMs with simple labels, the performance of the few-shot learning models demonstrated a slight improvement over zero-shot learning models, indicating that even a small number of labelled examples can significantly improve prediction performance. However, the performance of the fine-tuning models decreased slightly compared to the few-shot learning models. This may be due to the limited differences between the labelled examples used during fine-tuning, where false labels can lead to complementary datasets and reduced model performance.[34] To explore this further, we tested models on more complex labels designed to increase the differences in embedding value. The results revealed that neither zero-shot nor few-shot learning models enabled effective discrimination of the more complex categories. Thus, considering both annotation cost and training efficiency, a simple label schema combined with a few-shot learning model appears to be the most practical and effective strategy.

All three LLMs evaluated in the text information extraction task had F1 scores above 0.89 across all categories of ceramic materials, indicating that the fine-tuning model can efficiently extract structured data from abstracts when provided with standardized annotations. This suggests strong potential for applying the models to a broader range of data processing tasks and application scenarios. Compared with traditional manual extraction approaches reported in previous AI applications to dental materials, our LLM-based workflow markedly improves processing efficiency. The average time required to analyse a single article was reduced to approximately 1 minute, whereas the mean completion time for systematic reviews encompassing 195 articles has been reported as 67.3 weeks.[41] In addition, the screening scope expanded from several hundred publications to over ten thousand studies.[6,7] Together, these gains highlight substantial improvements in both efficiency and the breadth of evidence coverage. Furthermore, fine-tuning open-source LLMs requires far fewer training samples than BERT-based models

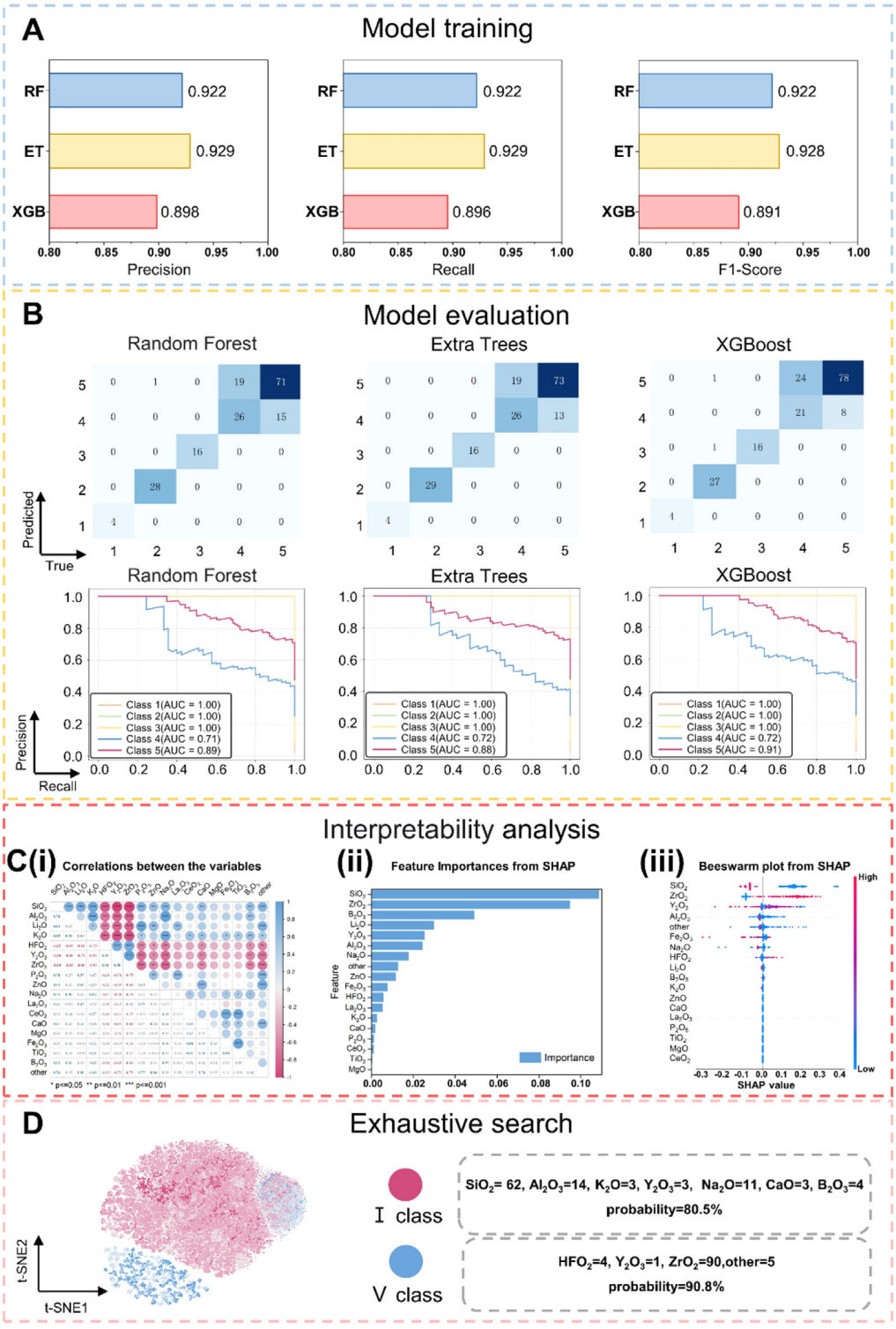


Figure 4 – Machine learning model development and evaluation. (A) Comparative classification metrics – including precision, recall, and F1 score – of the RF, ET, and XGB models. (B) Confusion matrices and precision-recall (PR) curves on the test set for all three models. (C) Correlation matrix of input variables (i), global feature importance based on SHAP values (ii), and SHAP beeswarm plot illustrating the positive or negative contribution of each feature to the prediction of Class V samples (iii). (D) T-distributed Stochastic Neighbor Embedding (t-SNE) plot showing the distribution of five types of ceramic materials in the virtually generated compositional space, along with the predicted compositions corresponding to the highest and lowest flexural strength.

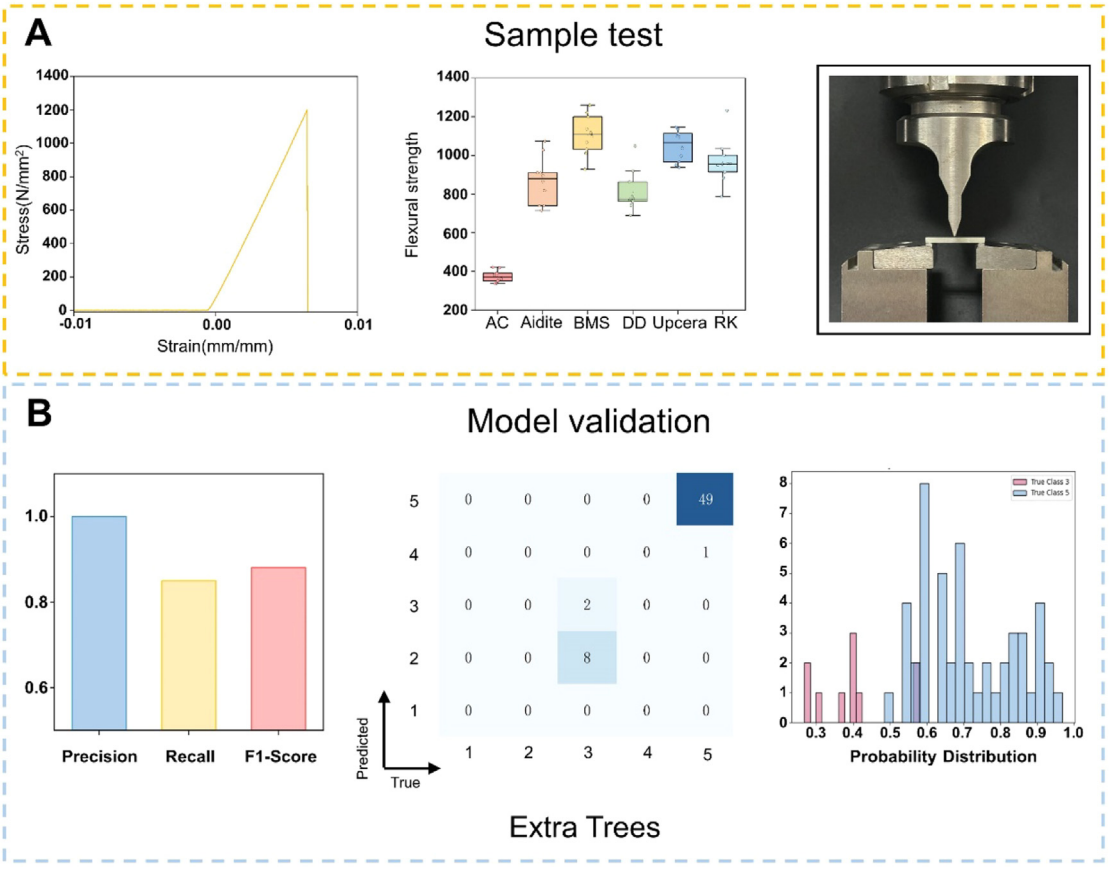


Figure 5 – Experimental validation of model predictive performance. (A) Stress–strain curves from three-point flexural strength tests and the distribution of sample measurements. (B) Classification performance on the external test set: precision, recall, and F1 score metrics (left), along with confusion matrix evaluation (middle) and predicted probability distributions (right).

to reach or exceed their performance level.[22,42] Beyond text, our framework innovatively incorporates multimodal data processing capabilities to extract structured information from tables, which offers enhanced interpretability compared to textual content alone. In addition, we fine-tuned the dental ceramic model using a small amount of annotated dental adhesive data and achieved high performance, indicating that even a limited set of domain-specific data can effectively adapt the model to new sub-domains. This highlights the transferability and broad applicability of our framework across diverse domains within dental materials research.

The ML model developed in this study not only demonstrated high accuracy in predicting the flexural strength of dental ceramic materials but also effectively identified optimal compositional combinations to achieve targeted mechanical properties. The feature importance analysis reveals that ZrO_2 and SiO_2 are the primary components influencing flexural strength. Integrated findings from SHAP analysis and exhaustive compositional searches indicated that higher ZrO_2 content positively contributes to flexural strength, whereas increased SiO_2 content is associated with reduced strength. These observations are supported by experimental evidence in the literature,[43] further validating the model's predictive capabilities and interpretability. Importantly, these insights extend beyond modelling performance: by identifying the key compositional drivers of strength and providing model-derived optimal formulations, the framework offers a data-driven basis for accelerating the formulation and optimization of new ceramic systems. At the clinical level, the ability to clearly indicate which components enhance or diminish mechanical reliability can help clinicians choose materials with superior performance and lower failure risk.

Dental clinical decision-making frequently requires weighing multiple factors, yet supporting evidence is often scattered across heterogeneous studies. The LLM-based framework proposed in this study helps bridge these methodological gaps by enabling scalable literature mining, efficient structuring of complex materials data, and downstream predictive modelling within a unified workflow. These capabilities are essential for synthesising fragmented evidence, accelerating materials innovation, and supporting evidence-based decision-making within the dental field.

As a first step in applying LLMs to dental materials research, there are still shortcomings in the current study. Although we retrieved literature through keyword searches and included the top 10,000+ most relevant articles, we did not apply additional filters such as 'highly cited publications' or 'SCI-indexed journals'. This lack of selective criteria may introduce uncertainty in the overall quality of the included literature. Future studies should consider defining more precise search parameters tailored to their specific research focus. Among the 219 records automatically extracted,

missing and duplicate entries were unavoidable, and the use of records still requires manual intervention for data cleaning. Despite the strong performance of LLMs in mining dental materials literature, their effectiveness ultimately depends on the quality of the retrieved training data and the reliability with which the models follow instructions. Addressing persistent issues in these data sources – such as bias, hallucinations, and domain mismatch – remains a critical area for further improvement.[44] Moreover, the application of LLMs in medical literature mining must be approached with caution; rigorous expert oversight is essential to ensure accuracy and to prevent biased or erroneous outputs.[45]

Furthermore, other important experimental parameters such as synthesis methods and processing conditions have not been included in the current information extraction pipeline. Future studies could enhance the comprehensiveness of the database by incorporating more detailed processing parameters and experimental conditions, and by representing components using chemical formulas instead of relying solely on chemical names. To enhance the clinical adaptability of predictive modelling, it is also crucial to integrate more clinically relevant simulation conditions. For example, incorporating flexural strength data obtained under aging conditions such as thermal cycling and mechanical loading, along with a broader range of performance metrics, could bridge the gap between laboratory predictions and clinical outcomes.[46,47] These improvements will contribute to a more comprehensive and clinically applicable database of dental ceramic materials. As the database continues to evolve to include richer properties, experimental parameters, and compositional information, it will enable the development of more robust ML models capable of more accurately predicting the long-term clinical performance of dental ceramics.

Conclusion

In summary, this study introduces for the first time an automated LLM-based workflow for scientific data extraction in dental materials research. By combining automated literature download, text classification, and extraction of both textual and tabular information, we generated a highly structured materials database. The entire data mining pipeline demonstrated reliable accuracy, highlighting the framework's potential for large-scale scientific data harvesting. Based on this, we then trained data-driven predictive models using the assembled database, enabling researchers and manufacturers to input experimental material compositions and directly predict their properties, thereby shortening development timelines. This work establishes a foundation for large-scale data processing and in-depth analysis in materials science, and provides an important reference for future research directions in predictive modelling and synthesis pathway optimization.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Houqi Zhou: Writing – original draft, Writing – review & editing, Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization. **Yaxin Bai:** Writing – original draft, Writing – review & editing, Conceptualization, Supervision, Visualization. **Yuan Chen:** Writing – review & editing, Formal analysis, Methodology, Software, Visualization. **Dongqi Fan:** Writing – review & editing, Methodology, Software, Supervision, Visualization. **Peng Wang:** Writing – review & editing, Data curation, Resources, Supervision, Validation. **Ping Ji:** Writing – review & editing, Conceptualization, Resources, Supervision. **Tao Chen:** Writing – review & editing, Conceptualization, Funding acquisition, Project administration, Resources, Software, Supervision, Visualization.

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Data availability

Data and source code can be obtained from the corresponding authors on reasonable request.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.identj.2025.109358.

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